Multilayer local score with an entropy-barrier window

Does a second, larger-window LS layer raise Inception Score?

Multilayer layer is dominated; widening LS windows x2 lifts CIFAR-10 IS 2.78 -> 6.29 with zero copies

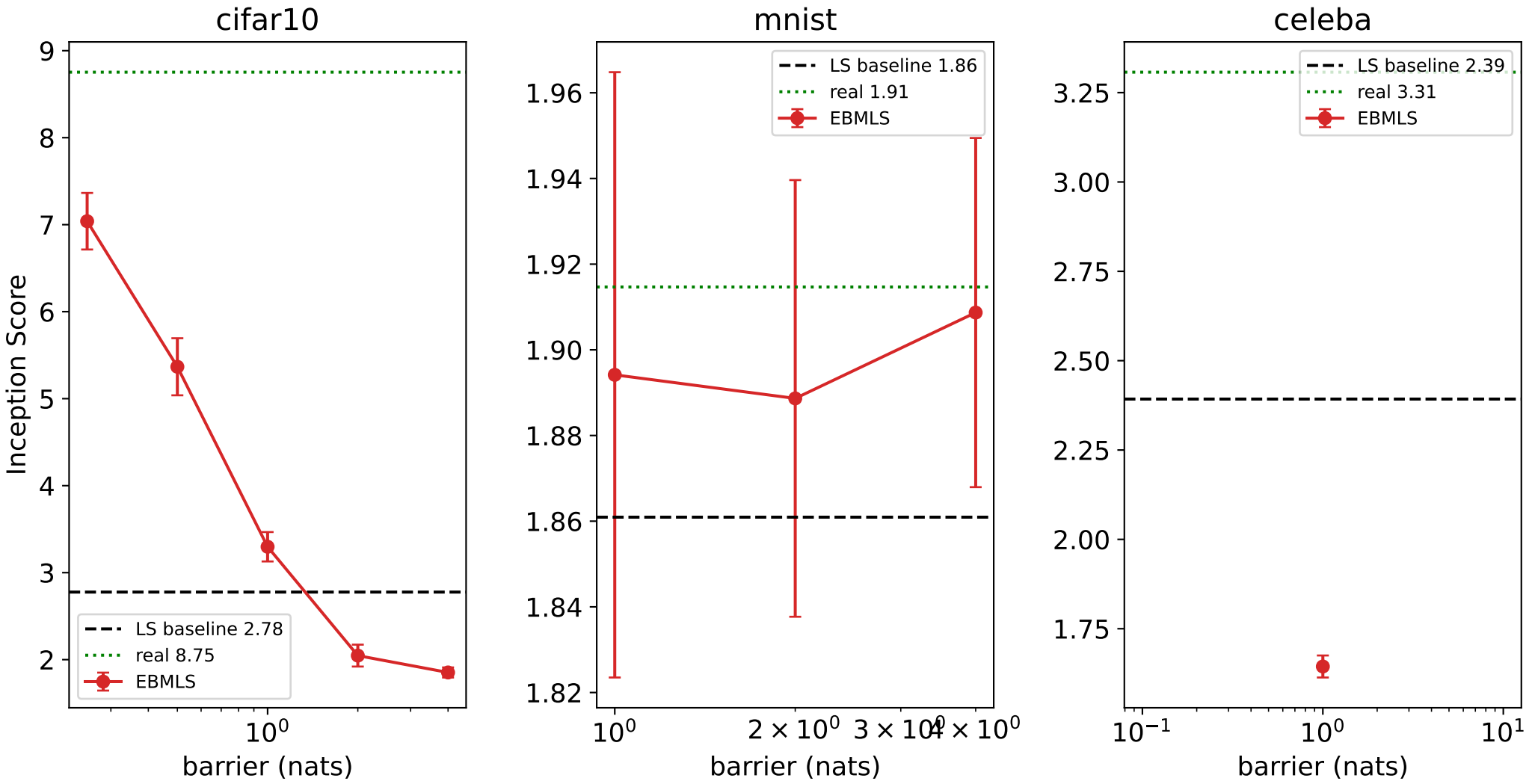
Layer 2 sees only layer 1's estimate, forms a posterior over all training images in a larger window, and adds the expected error of that estimate. The window at each step is the largest whose posterior entropy is still above a barrier — i.e. has not collapsed onto one image.

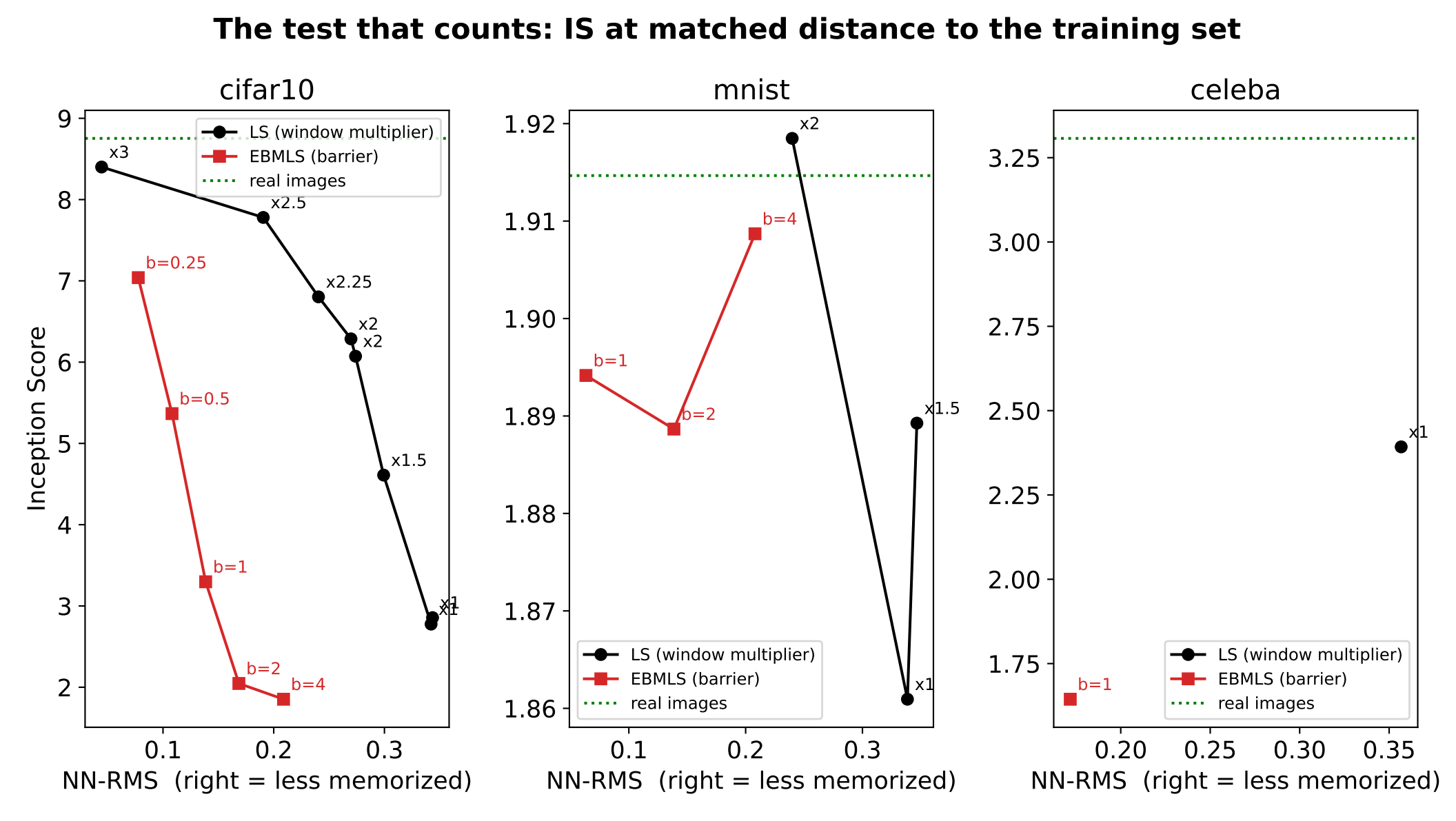
2026-10-21 · commit 2c2dc90 · seed 0 · 500 samples/config · IS over 5 splits

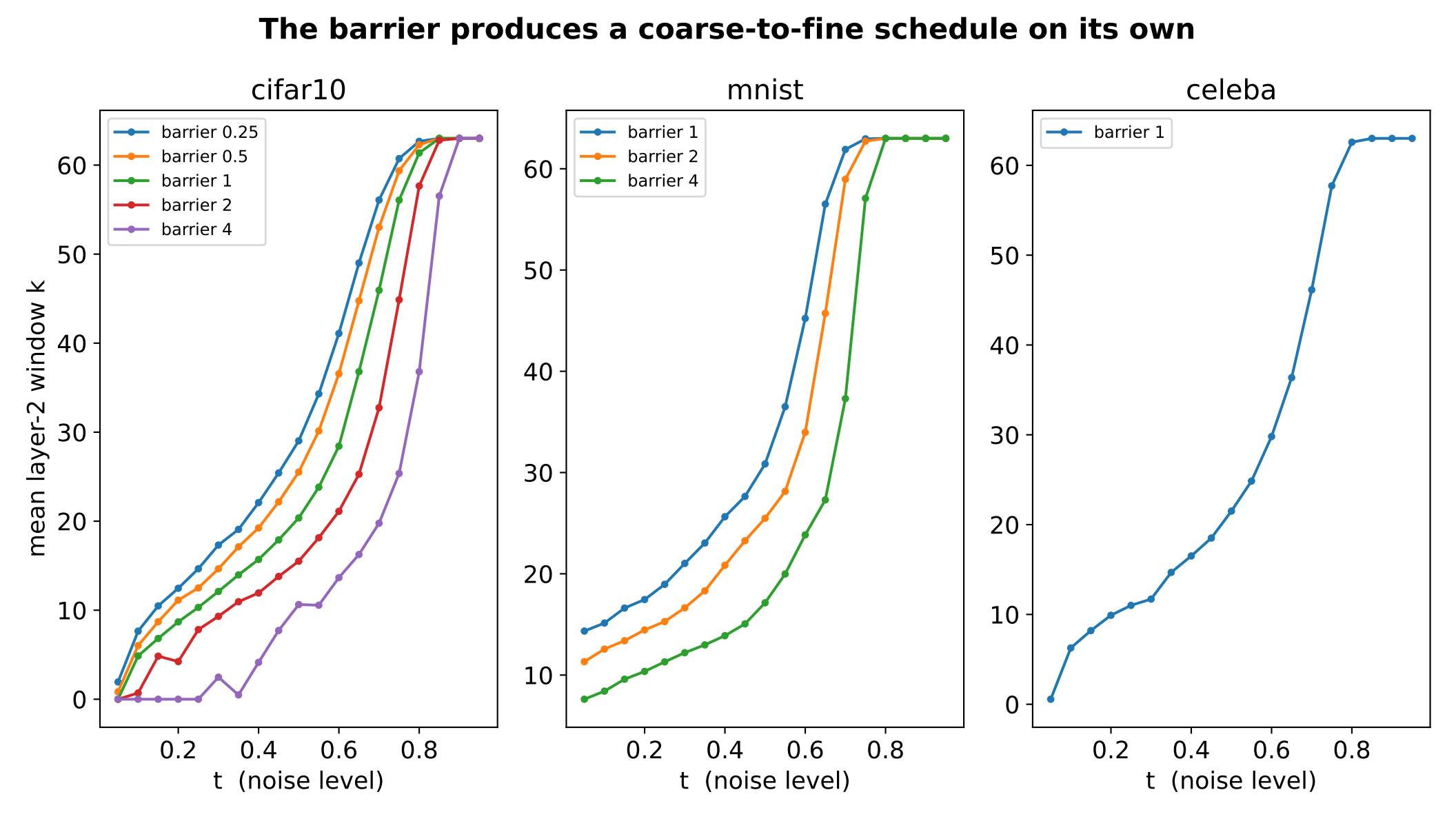
Metrics

NN-RMS: median distance to nearest training image — higher = less memorized

dataset	config	seed	IS	NN-RMS	copy frac
celeba	real images	0	3.307 ± 0.119	—	—
celeba	LS	0	2.393 ± 0.043	0.357	0.000
celeba	EBMLS barrier 1	0	1.644 ± 0.031	0.172	0.000
cifar10	real images	0	8.753 ± 0.465	—	—
cifar10	LS	0	2.777 ± 0.157	0.342	0.000
cifar10	LS	1	2.857 ± 0.061	0.343	0.000
cifar10	LS (windows x1.5)	0	4.610 ± 0.215	0.299	0.000
cifar10	LS (windows x2)	0	6.286 ± 0.563	0.270	0.000
cifar10	LS (windows x2)	1	6.073 ± 0.590	0.274	0.000
cifar10	LS (windows x2.25)	0	6.803 ± 0.365	0.240	0.028
cifar10	LS (windows x2.5)	0	7.780 ± 0.360	0.191	0.146
cifar10	LS (windows x3)	0	8.401 ± 0.467	0.045	0.514
cifar10	EBMLS barrier 0.25	0	7.040 ± 0.325	0.078	0.196
cifar10	EBMLS barrier 0.5	0	5.367 ± 0.328	0.108	0.058
cifar10	EBMLS barrier 1	0	3.298 ± 0.169	0.139	0.004
cifar10	EBMLS barrier 2	0	2.046 ± 0.126	0.169	0.000
cifar10	EBMLS barrier 4	0	1.853 ± 0.058	0.209	0.000
mnist	real images	0	1.915 ± 0.027	—	—
mnist	LS	0	1.861 ± 0.062	0.339	0.000
mnist	LS (windows x1.5)	0	1.889 ± 0.054	0.347	0.000
mnist	LS (windows x2)	0	1.918 ± 0.091	0.240	0.078
mnist	EBMLS barrier 1	0	1.894 ± 0.071	0.063	0.460
mnist	EBMLS barrier 2	0	1.889 ± 0.051	0.139	0.160
mnist	EBMLS barrier 4	0	1.909 ± 0.041	0.208	0.008

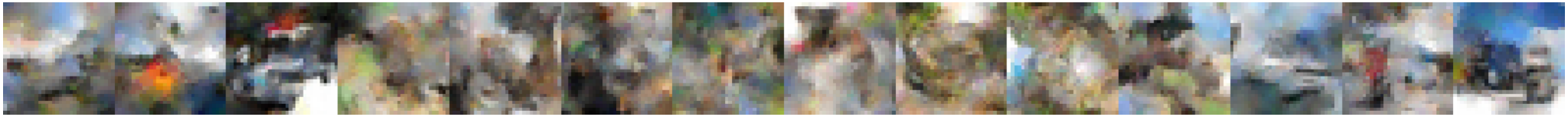

Inception Score vs entropy barrier
cifar10
mnist
celeba
Inception Score
barrier (nats)
barrier (nats)
barrier (nats)
LS baseline 2.78
real 8.75
EBMLS
LS baseline 1.86
real 1.91
EBMLS
LS baseline 2.39
real 3.31
EBMLS


The test that counts: IS at matched distance to the training set
cifar10
mnist
celeba
Inception Score
NN-RMS (right = less memorized)
NN-RMS (right = less memorized)
NN-RMS (right = less memorized)
LS (window multiplier)
EBMLS (barrier)
real images
x3
x2.5
x2.25
x2
x2
x1.5
x1
x1
b=0.25
b=0.5
b=1
b=2
b=4
x2
b=4
b=1
b=2
x1.5
x1
b=1
x1
LS (window multiplier)
EBMLS (barrier)
real images
LS (window multiplier)
EBMLS (barrier)
real images


The barrier produces a coarse-to-fine schedule on its own
cifar10
mnist
celeba
barrier 0.25
barrier 0.5
barrier 1
barrier 2
barrier 4
mean layer-2 window k
t (noise level)
0
10
20
30
40
50
60
0.2
0.4
0.6
0.8

cifar10 samples: baseline, best non-copying, best EBMLS, and the memorizing end

LS — IS 2.78, NN-RMS 0.342, copies 0.0%



LS (windows x2) — IS 6.29, NN-RMS 0.270, copies 0.0%



EBMLS barrier 0.25 — IS 7.04, NN-RMS 0.078, copies 19.6%



LS (windows x3) — IS 8.40, NN-RMS 0.045, copies 51.4%



Where this leaves us

1. The multilayer layer does not pay for itself.
   Both the entropy-barrier MLS and plain LS climb toward the real-image Inception Score as they drift toward the training set. At every matched nearest-neighbour distance, plain LS is the higher of the two, e.g.

   EBMLS barrier 0.25 : IS 7.04, 19.6% of samples are copies
   LS windows x2.5 : IS 7.78, 14.6% of samples are copies

   So the second layer buys Inception Score only by memorizing, and buys it less efficiently than simply widening the single-layer window. As specified, layer 2 replaces rather than refines: its weights sum to 1 at each pixel, so x_hat_1 + delta is exactly the layer-2 posterior mean over training images, with layer 1 entering only through the weights. That is what makes it collapse onto training images as soon as the window is allowed to grow.

2. Window width is where the Inception Score actually was.
   The scales file is far from the best operating point for this metric. Scaling every per-step window by 2 takes CIFAR-10 from IS 2.78 to 6.29 with no sample